KHOI NGUYEN

Electrical Engineering Student — Digital ASIC Design, RISC-V & AI Hardware

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Education

University of Manitoba **Aug 2023–Dec 2027**

Bachelor of Science in Electrical Engineering (GPA: 4.09/4.5) *Winnipeg, Canada*

Awards: Undergraduate Research Award (2025), Dean's Honour List (2023–2026), 3rd Place Undergraduate Research (2025)

Experience

University of Waterloo **April 2026–Sept 2026**

ASIC Design Co-op / Visiting Research Fellow – CMOS Design and Reliability Group *Waterloo, ON*

- Submitted a 2-core RV32IM test chip for fabrication in GF 12LP+ (12nm FinFET), with Calibre DRC/LVS clean. Owned my CPU block's full RTL-to-GDSII flow (Genus, Innovus, Joules, Modus).
- Architected 6 variable D-Q delay flip-flop fault-detection architectures with custom scan chain/DFF designs for delay-fault and timing-error detection.
- Ensured testability of the 12nm tape-out test chip with Cadence Modus DFT: scan insertion, ATPG and fault-simulation debug, reaching 97.5% stuck-at and 97.8–97.9% transition-fault coverage.

Nano-Systems Fabrication Laboratory (Class 10 Cleanroom) **May 2025–April 2026**

Student Researcher *University of Manitoba*

- Executed end-to-end CMOS and MEMS fabrication in a Class 10 cleanroom, including 200 nm DUV photolithography, ion implantation, RIE/ICP etching, and PVD/CVD thin-film deposition.
- Developed and optimized 200 nm DUV lithography processes (alignment, exposure dose, focus tuning), achieving high pattern fidelity and sub-micron critical dimension (CD) control.
- Optimized plasma etch and lithography recipes, reducing defect rate by 30–50% and improving etch throughput by 40%.

Projects

Adaptive DVFS RISC-V CPU with Timing-Error Detection: 12nm FinFET Tape-Out (GF 12LP+)

- Submitted the DRC/LVS-clean GDSII (11,506 µm², 21,892 instances) for fabrication on GlobalFoundries 12nm MPW run 2605DF (MPW1448) via CMC, Oct 2026; chips expected Aug 2027.
- Architected timing-error detection flip-flops to enable adaptive dynamic voltage-frequency scaling (DVFS), targeting energy efficiency gains under variable workload.
- Closed 1 GHz timing with zero violations across 4,739 paths (Genus, MMMC constraints, full LEF/technology-library integration) and completed floorplanning, power planning, placement and CTS in Innovus; post-route power 24.25 mW at 0.81 V (SS).

NVIDIA NVDLA AI Accelerator: Full-Chip Implementation in 12-nm FinFET | *Cadence Genus, Modus, Innovus*

- Closed 1.11 GHz timing (slow corner, 2 clock domains) on a 6.9M-instance NVDLA accelerator in 12nm FinFET, taking a 6-partition hierarchical flow with memory BIST from RTL to foundry-signoff GDSII
- Inserted integrated clock-gating cells (ICGs) to cut dynamic clock-tree power for a total power of 7.8 W.
- Generated 128 dual-port SRAM hard macros (RVT/MVT) with a foundry memory compiler, replacing behavioral RAM models to enable synthesis and P&R
- Completed sign-off placement, CTS and routing in Innovus at 62.75% cell density, with 0 DRC violations and LVS clean.

Clock-Gated int4 NPU (2x2 MAC Array), SKY130 Tiny Tapeout | *Verilog, cocotb, LibreLane, OpenSTA*

- Took a 2x2 int4 NPU (16-bit accumulators, clock gating, scan chain) from RTL to a signed-off SKY130 tile: 12-corner timing met at 50 MHz, DRC/LVS clean, TTSKY26d submission due Nov 2026.
- Clock gating cuts core power 45–59% (191 vs. 462 µW idle, pre-silicon estimate); verified with cocotb, SDF gate-level simulation and 26M-cycle RTL-vs-netlist co-simulation, 0 mismatches.

Electronics-Printed MEMS Lorentz Relay with Electrostatic Hold-Down | *COMSOL, Voltera NOVA, RIE*

- Designed and fabricated a hybrid MEMS relay (Lorentz-force actuation with electrostatic hold-down) on flexible polyimide using Voltera NOVA printed electronics and O_2 RIE release.

Technical Skills

Digital Design: Verilog, RISC-V, RTL, logic synthesis, STA, MMMC timing closure, clock gating, low-power flow
Physical Design & DFT: Cadence Genus, Innovus, Joules, Modus, Pegasus, Calibre DRC/LVS, Siemens Tessent MBIST, floorplanning, PnR, CTS, scan/ATPG
Custom IC & Fab: Virtuoso, ADE/Spectre, 12 FinFET, DRC/LVS, photolithography, ion implantation, CVD, RIE
Programming: Python, Tcl, MATLAB, Linux/Unix, ARM assembly

Certifications

Cadence Advanced Synthesis with Genus May 2026
Cadence Low-Power Flow with Innovus Implementation System May 2026